**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**

**Department of Computer Science and Engineering**

**CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis**

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|---|---|---|---|
| **Course Code:** | CSA10 / CSA1063 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO2 | **Assessment:** | Assessment Tool 2 – Scenario-Based Requirement Analysis |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO2 Statement:** | *Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills* | | |
| **Student Name:** | T. Niranjan Kumar | **Reg. No.:** | 192473007 |
| **Date:** | 3/08/2026 | **Duration:** | 60 minutes |

**Code of Conduct**

*I T. Niranjan Kumar (192473007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: T. Niranjan Kumar

## Annexure A – Scenario-Based Requirement Analysis

**Instructions to Students:**

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

### Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

### Rubrics for Calculation

| Criteria | Excellent (4 Marks) | Good (3 Marks) | Satisfactory (2 Marks) | Needs Improvement (1 Mark) |
|---|---|---|---|---|
| **Scenario Understanding and Problem Analysis (30%)** | Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis. | Demonstrates good understanding with minor omissions. | Basic understanding with limited analysis. | Poor understanding; major aspects missing or incorrect. |
| **Functional and Non-Functional Requirement Identification (25%)** | Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements. | Identifies most requirements with minor classification errors. | Identifies only basic requirements; several omissions. | Requirements are incomplete, inaccurate, or poorly classified. |
| **Actor Identification and Use Case Modelling (20%)** | Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards. | Most actors and use cases identified; diagram has minor errors. | Limited actors/use cases identified; diagram partially correct. | Actors/use cases missing or incorrect; diagram absent or inaccurate. |
| **Software Requirement Specification (SRS)** | SRS is well-structured, complete, professionally organized, and | SRS is organized with minor missing sections or formatting issues. | SRS includes only essential sections with limited detail. | SRS is incomplete, poorly organized, or |

# 1. INTRODUCTION

## 1.1 Overview

Open-source software has transformed the software development industry by enabling developers across the world to collaborate on real-world projects. Platforms such as GitHub provide access to millions of repositories where developers can contribute source code, fix bugs, improve documentation, and participate in community discussions. These contributions not only improve software quality but also help developers gain practical knowledge beyond academic learning.

For beginner developers, contributing to open-source projects is an excellent opportunity to improve programming skills, understand collaborative software development, and build an impressive technical portfolio. Many technology companies consider open-source contributions as valuable evidence of practical experience during recruitment. However, beginners often struggle to identify projects that match their knowledge level because of the enormous number of repositories and the varying complexity of issues available on GitHub.

The proposed **Open-Source Contribution Tracker and Recommendation System for Beginner Developers** addresses these challenges by providing personalized recommendations, centralized contribution tracking, portfolio generation, and skill progression analysis. The platform simplifies the process of discovering beginner-friendly projects and encourages continuous learning through structured guidance.

## 1.2 Purpose

The purpose of this Software Requirement Specification (SRS) document is to define the complete requirements of the proposed system in a structured and organized manner. It serves as a communication bridge between stakeholders, developers, testers, and project evaluators by clearly describing the objectives, features, constraints, interfaces, and expected behavior of the application.

This document also provides a strong foundation for software design, implementation, testing, deployment, and future maintenance. By documenting every requirement before development begins, the project team can reduce misunderstandings, avoid unnecessary changes, and ensure that the final system satisfies user expectations.

## 1.3 Need for the System

Although GitHub provides advanced search capabilities, it is primarily designed for experienced developers who already understand repository structures and contribution workflows. Beginner developers often spend a significant amount of time searching for suitable repositories, understanding contribution guidelines, and identifying issues appropriate for their skill level.

A dedicated recommendation and tracking platform is therefore required to simplify this process. The proposed system helps users discover repositories based on their programming languages and interests, monitors their contributions across multiple repositories, and presents their achievements through an interactive dashboard. This enables beginners to focus more on learning and contributing rather than manually searching for suitable opportunities.

**Table 1. Project Overview**

| **Attribute** | **Description** |
|---|---|
| Project Name | Open-Source Contribution Tracker and Recommendation System |
| Application Type | Web-Based Software Platform |
| Target Users | Beginner Developers |
| Domain | Software Engineering |
| External Platform | GitHub |
| Main Objective | Recommend projects and track contributions |
| Expected Benefit | Simplify beginner participation in open-source development |

## 2. PROBLEM DESCRIPTION

Open-source software development has become one of the most important components of modern software engineering. Thousands of organizations actively maintain repositories on GitHub, providing opportunities for developers to contribute to real-world projects. Despite these opportunities, beginners frequently encounter difficulties while selecting suitable repositories because of the large number of available projects and the differences in their complexity levels.

Most repositories contain advanced codebases that require significant programming experience. Although labels such as **"Good First Issue"** and **"Help Wanted"** exist, beginners must manually search through numerous repositories to find relevant issues. This process consumes valuable time and often results in frustration when selected projects do not match the developer's skill level.

Another important challenge is the lack of a centralized platform that tracks contributions across multiple repositories. Beginners find it difficult to monitor pull requests, issue status, merged contributions, repository participation, and technology exposure in a single place. As a result, maintaining a professional portfolio becomes challenging.

The proposed system solves these challenges by integrating GitHub APIs to provide personalized project recommendations, beginner-friendly issue suggestions, contribution tracking, skill progression analysis, portfolio generation, and community insights. The

platform reduces the entry barrier for new contributors and encourages continuous participation in open-source development.

**Table 2. Existing System vs Proposed System**

| Existing System | Proposed System |
|---|---|
| Manual repository search | Intelligent repository recommendation |
| Generic GitHub search filters | Personalized recommendations based on skills |
| No centralized contribution tracking | Complete contribution tracking dashboard |
| Manual portfolio preparation | Automatic portfolio generation |
| Difficult project selection | Beginner-friendly project suggestions |
| Limited beginner guidance | Skill-based recommendations and notifications |

## 3. SCOPE OF THE SYSTEM

The scope of the proposed system is to provide a comprehensive platform that assists beginner developers in identifying suitable open-source projects, managing their contribution activities, and monitoring their professional growth. The system integrates with GitHub through secure APIs to retrieve repository information, issue labels, pull request status, and contribution history.

Users can create their profiles by specifying programming languages, technical interests, and preferred technology domains. Based on this information, the recommendation engine identifies repositories that match the user's profile and highlights beginner-friendly issues such as **Good First Issue** and **Help Wanted**. This significantly reduces the time required to discover appropriate projects.

The system also provides a centralized dashboard that displays contribution statistics, merged pull requests, issue participation, programming language diversity, and repository activity. Through graphical reports and progress indicators, users can evaluate their learning journey and identify areas for improvement. An automated portfolio generation feature summarizes contributions in a professional format suitable for resumes and job applications.

Furthermore, the platform promotes active participation in open-source communities by providing notifications whenever new beginner-friendly issues become available. Community insights such as maintainer responsiveness, mentorship availability, and contributor-friendliness help users choose projects that offer a positive learning experience.

Although the system is primarily designed for beginner developers, its modular architecture allows future expansion to support intermediate and advanced contributors through AI-

powered recommendations, predictive analytics, and advanced contribution scoring mechanisms.

## 4. FUNCTIONAL REQUIREMENTS

Functional requirements describe the core operations that the proposed system must perform to satisfy user needs. These requirements define how the application interacts with beginner developers, administrators, and external services such as GitHub. The primary objective is to simplify the process of discovering suitable open-source projects while providing an organized environment to monitor contribution activities.

The system shall provide a secure registration and authentication mechanism that allows users to create accounts and connect their GitHub profiles. After successful authentication, users can enter their programming languages, technical interests, and preferred development domains. Based on this information, the recommendation engine will analyze available repositories and identify projects that best match the user's profile.

Another important functionality of the application is centralized contribution tracking. The platform shall automatically retrieve information about pull requests, commits, issues, merged contributions, and repository participation using GitHub APIs. Instead of visiting multiple repositories individually, users can monitor all contribution activities through a single dashboard.

The application shall also generate a professional portfolio that summarizes user contributions, programming languages used, repository participation, and project achievements. This portfolio can be used for internship applications, placement opportunities, and professional networking. Additionally, the notification system will continuously monitor repositories and alert users whenever new beginner-friendly issues become available that match their skill profile.

**Table 2. Functional Requirements**

| Requirement ID | Description | Priority |
|---|---|---|
| FR-01 | User Registration and Login | High |
| FR-02 | GitHub Account Integration | High |
| FR-03 | Skill Profile Management | High |
| FR-04 | Repository Recommendation | High |
| FR-05 | Beginner-Friendly Issue Recommendation | High |
| FR-06 | Contribution Tracking Dashboard | High |

| | | |
|---|---|---|
| FR-07 | Pull Request and Issue Monitoring | High |
| FR-08 | Portfolio Generation | High |
| FR-09 | Notification System | Medium |
| FR-10 | Community Insight Analysis | Medium |

## 5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality standards that ensure the proposed system performs efficiently, securely, and reliably. Unlike functional requirements, these requirements focus on system performance, usability, maintainability, security, scalability, and availability rather than specific system operations.

The application should respond quickly while retrieving repository information from GitHub APIs. Most user requests, including repository recommendations and dashboard loading, should be completed within a reasonable response time to provide a smooth user experience. The platform must also remain operational even when multiple users access the system simultaneously.

Security is another critical aspect of the proposed application. User credentials and GitHub authentication tokens should be protected using secure authentication mechanisms and encrypted communication. Access to personalized dashboards and contribution information must be restricted to authenticated users only.

The application should provide a simple and user-friendly interface suitable for beginner developers who may have limited experience with GitHub workflows. Navigation should be intuitive, and important features such as repository recommendations, contribution tracking, and portfolio generation should be easily accessible.

The software architecture should also support future scalability. As the number of users and repositories increases, the application should continue to deliver reliable performance without significant degradation. Furthermore, the modular design of the system should simplify future maintenance, updates, and feature enhancements.

## 6. USE CASE DESCRIPTIONS

A use case represents the interaction between a user and the proposed system to accomplish a specific task. In this project, the primary user is the beginner developer, while the administrator manages system operations. GitHub acts as an external system that supplies repository information, contribution history, and issue details through APIs.

The most common interaction begins when a new user registers and logs into the application. After connecting the GitHub account and selecting programming skills, the recommendation engine analyzes repository information and suggests suitable beginner-friendly projects.

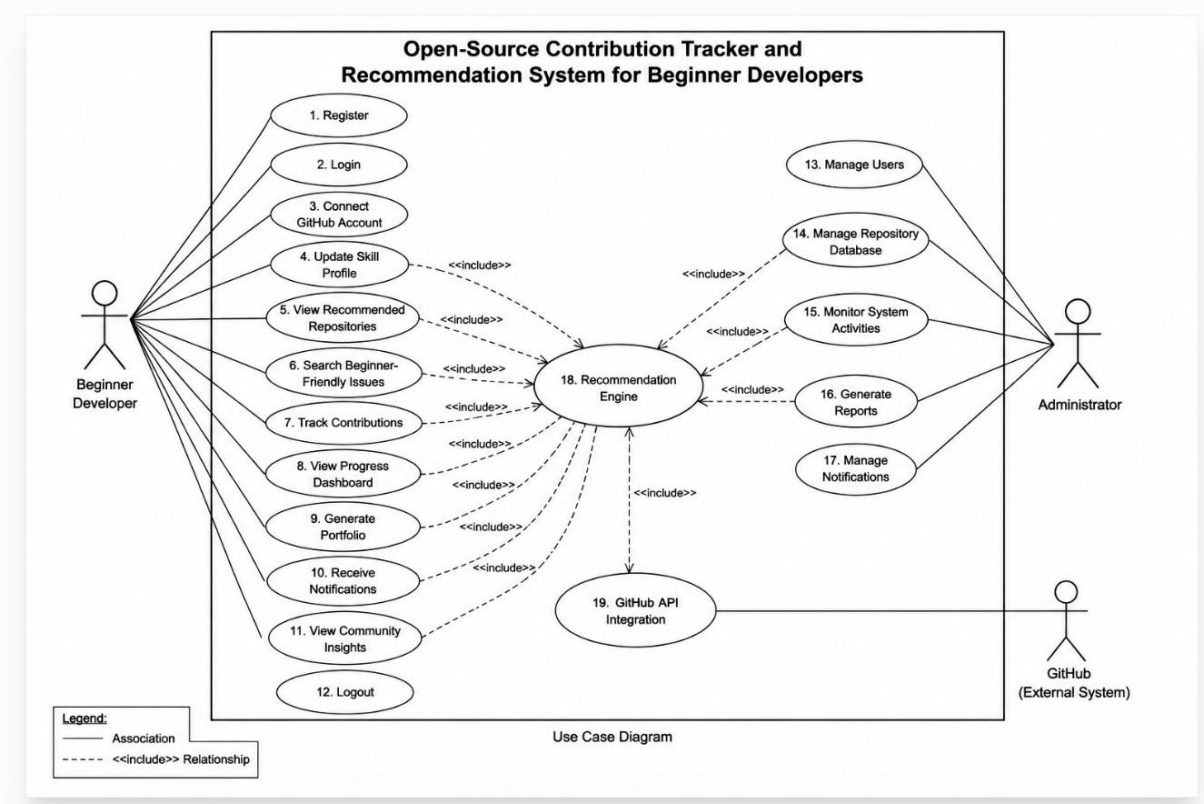
Users can browse recommended repositories, view available issues, participate in projects, and monitor contribution progress through the dashboard.

The application continuously synchronizes with GitHub to update pull request status, issue participation, merged contributions, and repository activity. Users can generate a professional portfolio at any time, while the notification service informs them whenever new beginner-friendly opportunities become available.

**Major Use Cases**

- User Registration
- User Login
- GitHub Account Integration
- Update Skill Profile
- View Recommended Repositories
- Search Beginner-Friendly Issues
- Track Contributions
- View Progress Dashboard
- Generate Portfolio
- Receive Notifications
- View Community Insights
- Logout

**Figure 1. Use Case Diagram**




**Use Case – Repository Recommendation**

**Primary Actor:** Beginner Developer

**Precondition:**
The user must be logged into the system and have completed the skill profile by selecting known programming languages and areas of interest.

**Main Flow:**

1. The user opens the Recommendation Dashboard.
2. The system retrieves repository information from GitHub.
3. The recommendation engine analyzes the user's skill profile.
4. Beginner-friendly repositories are filtered based on matching technologies and issue labels.
5. The recommended repositories are displayed to the user in ranked order.

**Postcondition:**
The user receives a personalized list of repositories that match their programming skills and learning objectives, enabling them to begin contributing more effectively.

## 7. DATA REQUIREMENTS

The proposed system requires well-organized and secure data management to provide accurate repository recommendations, maintain contribution history, and generate meaningful analytics. The database stores both user-specific information and repository-related details retrieved through GitHub APIs. Proper data management ensures efficient retrieval of information while maintaining data integrity and consistency.

User-related information includes personal details, GitHub profile data, programming skills, preferred technologies, contribution history, notification preferences, and generated portfolios. Repository-related information includes repository names, programming languages, issue labels, contributor statistics, pull request status, and project activity.

The system also maintains records of notifications, portfolio generation history, contribution summaries, and dashboard statistics. Sensitive information such as authentication credentials and access tokens is protected using secure storage mechanisms. Proper database normalization techniques are adopted to eliminate redundancy and improve system performance.

The database is designed to support future enhancements without affecting the existing application structure. Additional entities such as machine learning recommendation history, achievement badges, and mentor interactions can easily be incorporated through modular database expansion.

**Table 3. Main Database Entities**

| Entity Name | Purpose |
|---|---|
| User | Stores user profile and login information |
| Skills | Stores programming languages and interests |
| Repository | Stores recommended repository details |
| Issues | Maintains beginner-friendly issue information |
| Contributions | Tracks commits, pull requests, and issues |
| Notifications | Stores alerts sent to users |
| Portfolio | Maintains generated portfolio information |

## 8. EXTERNAL INTERFACE REQUIREMENTS

The proposed application interacts with users and external platforms through well-defined interfaces that provide a simple and efficient user experience. Since the system is primarily designed for beginner developers, the interface focuses on simplicity, clarity, and ease of navigation.

The user interface consists of responsive web pages that allow users to register, log in, connect GitHub accounts, update skill profiles, browse recommended repositories, monitor contributions, and generate portfolios. Each page follows a consistent layout with intuitive navigation menus, dashboards, and interactive charts.

The system communicates with GitHub through official REST APIs. These APIs provide repository information, issue labels, pull request status, contribution history, and repository statistics. Secure authentication mechanisms ensure that user data is exchanged safely between the application and GitHub.

The application is designed to operate on modern web browsers including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. It is also compatible with desktop computers, laptops, tablets, and mobile devices, allowing users to access the platform from different environments without compromising usability.

**User Interface**

The user interface should be clean, responsive, and beginner-friendly. Important functions such as repository recommendations, dashboard reports, contribution tracking, and portfolio generation should be accessible within a few clicks. Visual consistency, readable typography, and intuitive navigation contribute to a positive user experience.

**Software Interface**

The software interface is established through GitHub REST APIs, which provide repository information, pull request details, issue labels, contributor statistics, and user activity. API integration ensures that recommendations remain updated with current repository information.

**Hardware Interface**

The application requires a standard desktop computer, laptop, tablet, or smartphone with an internet connection. No specialized hardware components are required because the platform is entirely web-based.

**Communication Interface**

The application communicates securely over HTTPS protocols to ensure confidentiality and integrity of transmitted information. Email notifications and GitHub authentication services are also utilized whenever required.

## 9. SYSTEM FEATURES

The proposed platform combines recommendation, tracking, visualization, and portfolio generation features into a single integrated application. Each feature has been designed specifically to simplify the open-source contribution journey for beginner developers.

The recommendation engine analyzes programming languages, technology interests, and previous contribution history to suggest repositories containing beginner-friendly issues. This personalized approach significantly reduces the effort required to discover suitable projects and improves the probability of successful first contributions.

The contribution tracker automatically records commits, pull requests, merged requests, issue participation, and repository activity. These details are presented through an interactive dashboard that enables users to monitor their learning progress and evaluate personal growth over time.

Another important feature is automated portfolio generation. Instead of manually collecting contribution details, users can generate a professional portfolio that summarizes repositories, programming languages, achievements, contribution statistics, and project participation. This portfolio can be used for internships, placement interviews, scholarships, and professional networking.

The notification module continuously monitors repositories for new beginner-friendly issues that match the user's programming profile. Timely notifications help users identify opportunities before they become unavailable. Community insights further improve decision-making by displaying maintainer responsiveness, contributor friendliness, and mentorship availability for each repository.

Overall, the proposed system integrates multiple essential services into a single platform, making open-source participation easier, more organized, and more productive for beginner developers.

## 10. ASSUMPTIONS

The development of the proposed system is based on several practical assumptions that simplify implementation while ensuring reliable system performance. These assumptions define the expected operating environment and user behavior during the execution of the application.

It is assumed that every user possesses a valid GitHub account and grants the necessary permissions required for accessing repository information and contribution history. Without proper authorization, the recommendation engine and contribution tracker cannot retrieve accurate repository data.

The system also assumes that users provide correct programming language preferences and technology interests during profile creation. Accurate user information allows the recommendation engine to generate relevant project suggestions and improve recommendation quality.

It is further assumed that GitHub APIs remain available and continue to provide repository information, issue labels, pull request details, and contribution history without significant changes to their functionality. Stable internet connectivity is also assumed because the application depends on real-time communication with GitHub services.

The proposed application assumes that repository maintainers correctly assign labels such as **"Good First Issue"** and **"Help Wanted"** so that beginner-friendly repositories can be identified accurately. Proper labeling significantly improves recommendation quality and user satisfaction.

## 11. CONSTRAINTS

Every software system operates under certain limitations that influence design decisions, implementation strategies, and overall system performance. The proposed application is also subject to several technical, operational, economic, legal, and time-related constraints.

One major technical constraint is the dependency on GitHub REST APIs. Any changes in API structure, authentication mechanisms, or usage limitations may directly affect repository recommendations and contribution tracking features. API rate limits imposed by GitHub may also restrict the number of requests that can be processed within a specific time period.

Operationally, the application requires continuous internet connectivity because repository information is retrieved dynamically from GitHub servers. Without network connectivity, users cannot receive updated recommendations or synchronize contribution history.

Economic constraints include cloud hosting expenses, database maintenance costs, and infrastructure resources required to support increasing numbers of users. Although the application primarily relies on publicly available GitHub APIs, long-term deployment may require additional server resources for improved scalability.

Legal constraints involve compliance with GitHub's API policies, privacy regulations, and data protection standards. The application must ensure secure handling of user credentials and personal information while respecting the terms and conditions established by external service providers.

Time constraints relate to software development, testing, deployment, and future maintenance schedules. Timely completion of project milestones is essential for successful implementation without affecting software quality.

**Table 4. Project Constraints**

| Constraint Type | Description |
|---|---|
| Technical | Dependency on GitHub REST APIs and internet connectivity |
| Operational | Requires active GitHub account and valid user profile |

| | |
|---|---|
| Economic | Cloud hosting and database maintenance costs |
| Legal | Compliance with GitHub API policies and privacy regulations |
| Time | Limited development and testing schedule |

## 12. REQUIREMENT VALIDATION

Requirement validation is one of the most important activities in software engineering because it ensures that the documented requirements accurately represent stakeholder expectations and project objectives. Validation helps identify ambiguities, inconsistencies, missing information, and conflicting requirements before software development begins.

The functional requirements have been reviewed to confirm that every major feature requested in the project objectives has been included within the Software Requirement Specification. User registration, GitHub integration, repository recommendation, contribution tracking, notification services, dashboard visualization, and portfolio generation are all represented clearly within the proposed system.

Similarly, the non-functional requirements have been validated to ensure acceptable standards of security, reliability, performance, scalability, maintainability, usability, and availability. These quality attributes guarantee that the application provides a stable and efficient experience for beginner developers.

Actors, use cases, database requirements, external interfaces, assumptions, and constraints have also been reviewed to verify consistency throughout the document. No conflicting requirements have been identified, and each functional component directly supports one or more project objectives. The resulting specification provides a complete foundation for software design and implementation.

The requirement validation process confirms that the proposed system satisfies the needs of beginner developers by providing an integrated platform for discovering open-source projects, monitoring contribution activities, and demonstrating professional growth through measurable contribution statistics and portfolio generation.

## 13. SYSTEM ARCHITECTURE

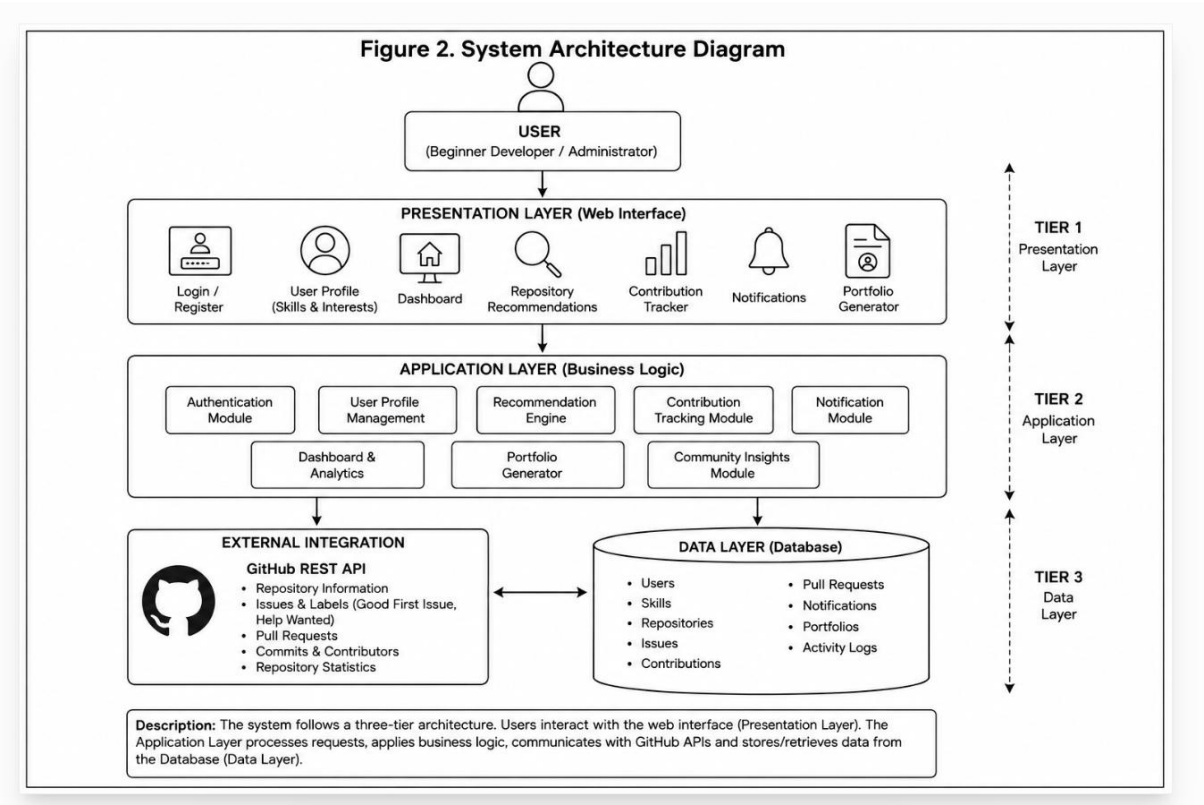
The proposed system follows a **three-tier architecture**, consisting of the Presentation Layer, Application Layer, and Data Layer. This architecture improves scalability, maintainability, and overall system performance by separating the user interface, business logic, and database components.

The **Presentation Layer** provides an interactive web interface through which users can register, log in, connect their GitHub account, browse recommended repositories, monitor contributions, and generate portfolios. This layer communicates directly with the application server through secure HTTP requests.

The **Application Layer** contains the core business logic of the system. It manages user authentication, repository recommendation, contribution tracking, notification services, dashboard generation, and portfolio creation. The Recommendation Engine processes user skills and repository information to generate personalized suggestions.

The **Data Layer** stores user profiles, programming skills, repository information, contribution history, notifications, and generated portfolios. The system also communicates with the GitHub REST API to retrieve real-time repository information, issue labels, pull request status, and contribution history.

**Figure 2. System Architecture Diagram**



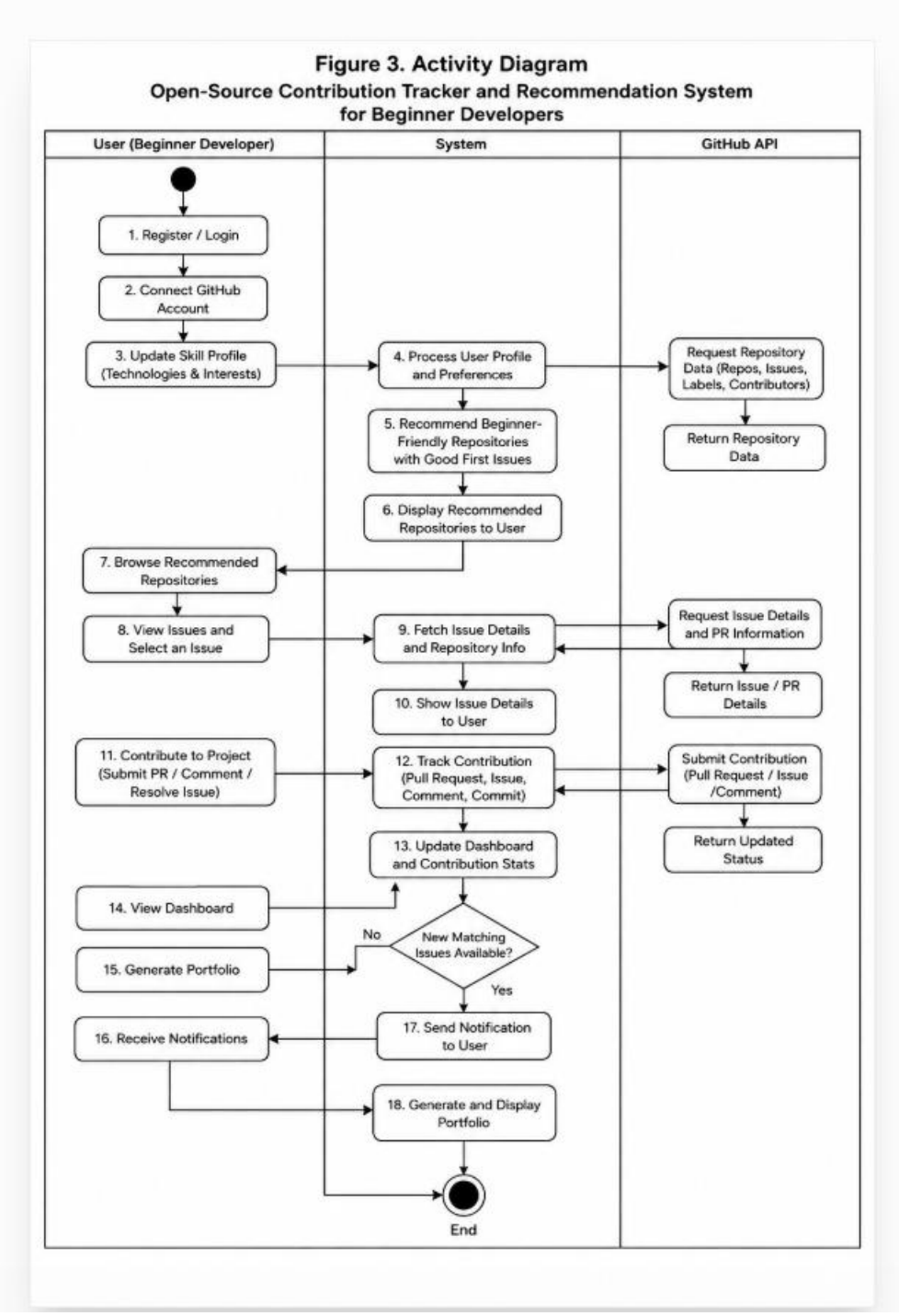

## 14. ACTIVITY DIAGRAM

The activity diagram illustrates the sequence of operations performed by the system from user authentication to repository recommendation and contribution tracking. It represents the workflow followed by a beginner developer while interacting with the application.

Initially, the user registers and logs into the system. After successful authentication, the user connects the GitHub account and selects programming skills and preferred technologies. The recommendation engine retrieves repository information through GitHub APIs and filters beginner-friendly repositories based on the user's profile.

The user can browse recommended repositories, participate in open-source projects, submit pull requests, and monitor contribution progress through the dashboard. Whenever new matching repositories become available, the notification module alerts the user. Finally, the system generates an updated professional portfolio summarizing all contribution activities.

**Figure 3. Activity Diagram**




## 15. ENTITY RELATIONSHIP (ER) DIAGRAM

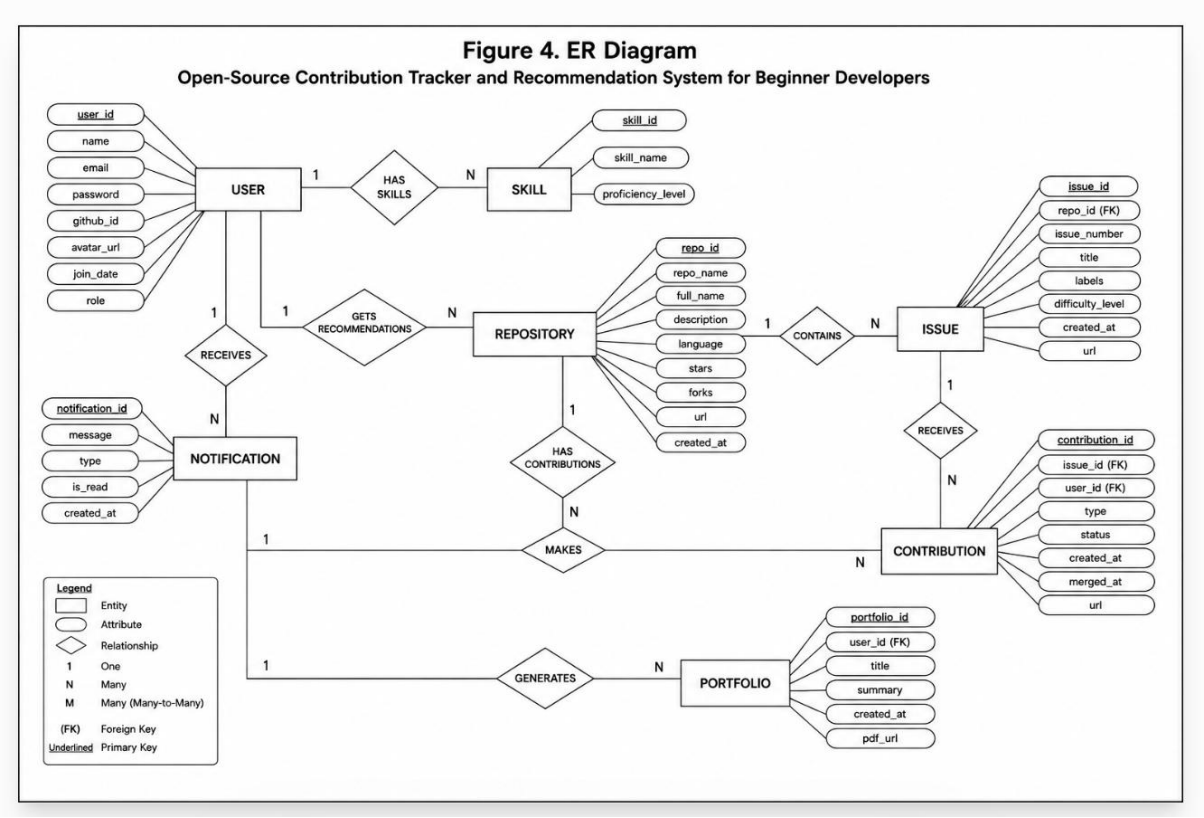
The Entity Relationship Diagram (ER Diagram) represents the logical database structure of the proposed application. It illustrates the relationships among the primary entities used by the system.

The **User** entity stores personal information and authentication details. Each user possesses one or more programming skills and receives multiple repository recommendations. Repositories contain multiple beginner-friendly issues, while each repository may receive

contributions from several users. Contribution records maintain pull request status, commit history, and issue participation. Notifications are generated for users whenever suitable repositories become available, and portfolio information summarizes contribution achievements.

This database structure minimizes redundancy while maintaining data consistency and efficient information retrieval.

**Figure 4. ER Diagram**




## 16. CONCLUSION

The **Open-Source Contribution Tracker and Recommendation System for Beginner Developers** provides an effective solution to the challenges faced by newcomers entering the open-source software ecosystem. By combining personalized repository recommendations, centralized contribution tracking, automated portfolio generation, and community insights within a single platform, the system significantly reduces the effort required to discover suitable projects and monitor contribution progress.

The proposed application enables beginner developers to build practical programming experience while improving technical confidence and professional visibility. Integration with GitHub APIs ensures that repository information remains accurate and up to date, while interactive dashboards help users monitor learning progress and contribution statistics effectively.

The modular architecture adopted by the system supports future enhancements such as artificial intelligence-based recommendation techniques, advanced contribution analytics, and

expanded support for intermediate and experienced developers. Overall, the proposed platform promotes continuous learning, active community participation, and long-term career development within the global open-source ecosystem.

## References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4. **[4]** Sharma, A., Kumar, P., & Singh, R. (2024). *Recommendation Systems for Open-Source Software Contribution Using GitHub Data.* **International Journal of Computer Applications (IJCA).**
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6. These **5 recent references (2022–2024)** are sufficient for the references section of your SRS report and look professional.